Common Lab Mistakes

EE5203 Microprocessors — what goes wrong, what you see, and how to fix it

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Almost every lost laboratory hour comes from one of the mistakes below. None of them is a coding mistake. They are all about **which folder you opened**, **where the `.vscode` folder sits**, and **which button you pressed**.

Use this page in two ways: read it once before Lab 1, and search it by symptom when something fails.

Check these three things first

Before you ask for help, look at these three:

1. **Title bar of VS Code.** It must show your project folder — the folder that contains the `.ioc` file. Not the folder above it.
2. **Explorer panel, top level.** You must see `.vscode`, `Core`, `Drivers`, `CMakeLists.txt`, `<name>.ioc` side by side.
3. **`.vscode` contents.** Five files: `tasks.json`, `launch.json`, `settings.json`, `fix_ld.cmake`, `STM32F401RETx_FLASH.ld`.

If all three are correct, the problem is in your code and you can debug it. If any one is wrong, fix it first — the error message you are reading is probably a side effect.

Symptom lookup

What you see	Go to
F5 asks you to "Select debugger" / "Select environment"	1, 4, 5
Ctrl+Shift+B says "No build task to run found"	1, 4, 5
`CMake Error: Could not read presets` / "No such preset"	1, 8
`STM32F401RETx_FLASH.ld:56: syntax error`	12, 5
`Unable to start debugging. Program file ...` `.elf does not exist`	7, 13
`CMake Error: The current CMakeCache.txt directory ... is different`	7
`undefined reference to '...'`	17
`cmake -P ... fix_ld.cmake` → file not found	4, 5
Compiler errors about `iostream`, `cl.exe`, `g++`	14
ST-LINK GDB server error, target busy, cannot connect	15
CubeMX keeps offering to download STM32CubeF4	10, 11
Your code is gone after you changed a pin in CubeMX	18

Opening the project

1. You opened the folder *above* the project

What you did: File → Open Folder → `C:\Work\2400001234`, the folder that holds all your projects.

What you see: No preset question. Ctrl+Shift+B reports **"No build task to run found"**. F5 asks you to pick a debugger. CMake Tools cannot find `CMakePresets.json`.

Why: VS Code reads `.vscode`, `CMakeLists.txt` and `CMakePresets.json` only at the **top level of the folder you opened**. One level up, they are invisible to it.

Fix: File → Open Folder → the folder that contains the `.ioc` file, for example `C:\Work\2400001234\Blink`. Check the title bar.

2. You opened a folder *inside* the project

What you did: Opened `Core`, `Core\Src`, or `build`.

What you see: The same symptoms as mistake 1. `main.c` opens and edits normally, which makes the mistake easy to miss.

Why: Same reason. The project files are now *above* the folder you opened.

Fix: Open the project folder itself. Being able to edit `main.c` proves nothing — any folder lets you edit files.

3. You opened a file, not a folder

What you did: Double-clicked `main.c`, or used File → Open File.

What you see: VS Code says **"You have not yet opened a folder"**. Nothing builds.

Why: Without an open folder there is no workspace, so there are no tasks and no debug configuration.

Fix: File → Open Folder. On the laboratory PCs, start from the **VS Code - EE Labs** shortcut and open the folder from there.

The `.vscode` template

4. You copied the five files instead of the folder

What you did: Opened `C:\Users\Public\EE5203_template\.vscode`, selected the files inside, and pasted them into your project.

What you see: `tasks.json`, `launch.json` and the rest sit loose next to `main.c` and `CMakeLists.txt`. Ctrl+Shift+B finds no task; F5 asks for a debugger.

Why: VS Code only reads these files from a folder named exactly `.vscode`. Loose in the project root they mean nothing.

Fix: Go one level up, copy the **folder** `.vscode` itself, and paste it into your project folder. Then delete the loose copies.

5. The `.vscode` folder is in the wrong place

What you did: Pasted `.vscode` into `C:\Work\2400001234` (next to the project), or into `Core`, or into `build`.

What you see: A mixture: no build task, or the build runs but stops at `STM32F401RETx_FLASH.ld:56: syntax error`, or cmake reports it cannot find `fix_ld.cmake`.

Why: The repair script looks for the linker script in the folder directly above itself. Put one folder off, it repairs the wrong folder — or nothing.

Fix: `.vscode` belongs **inside the project folder, next to the `.ioc` file**:

```
C:\Work\2400001234\Blink\
    .vscode\            <- here
    Core\
    Drivers\
    CMakeLists.txt
    Blink.ioc
```

6. You renamed `.vscode`

What you did: Windows hid the leading dot, or you saved it as `vscode`, `.vscode (2)`, or `vscode-template`.

What you see: As if the folder were missing.

Fix: The name must be exactly `.vscode`, with the dot, no space, no number.

7. You renamed or copied a project folder and changed nothing else

What you did: Copied `EE5203_L01_2400001234` and renamed the copy `EE5203_L02_2400001234` — or renamed a CubeMX project folder by hand.

What you see: One of two errors:

- `CMake Error: The current CMakeCache.txt directory ... is different than the directory ... where CMakeCache.txt was created` — the copied `build` folder still points to the old folder.
- The build succeeds, then F5 fails with **"Program file does not exist"** — it is looking for `build\Debug\EE5203_L02_2400` while the build produced `EE5203_L01_2400001234.elf`.

Why: The debug configuration takes the program name from the **folder name**. CMake takes it from one line in `CMakeLists.txt`. Renaming the folder changes only the first.

Fix: After every copy and rename:

1. delete the `build` folder in the new copy;
2. in `CMakeLists.txt`, set the project name to the new folder name:

```
set(CMAKE_PROJECT_NAME EE5203_L02_2400001234)
```

The Lab 2 and Lab 3 guides show these steps in Checkpoint A.

Creating the project in CubeMX

8. Toolchain/IDE is not CMake

What you did: Left Project Manager → Toolchain/IDE at **STM32CubeIDE**, or chose Makefile or EWARM.

What you see: There is no `CMakePresets.json` and no usable `CMakeLists.txt`. VS Code reports it cannot read presets, and nothing builds.

Fix: Project Manager → Toolchain/IDE → **CMake** → Generate Code again. Generate into a **new, empty folder**; do not mix two toolchains in one folder.

9. Bad project location or project name

What you did: Saved to the Desktop, to `H:`, to a OneDrive folder, or used a name with spaces or Turkish characters (`Ölçüm 1`, `Deney Bir`).

What you see: Unpredictable build failures, paths reported as not found, very slow builds, or files that vanish between sessions.

Why: The build tools pass these paths to the compiler as plain text. Spaces and non-ASCII characters break them, and network drives are too slow and too unreliable for a build.

Fix: Generate into `C:\Work\<studentnumber>`. Project names: letters, digits and underscore only.

10. You accepted the STM32CubeF4 download

What you did: Clicked **Yes / Download** when CubeMX offered the STM32CubeF4 package.

What you see: A long download that may fail, and a second copy of the package in your own user folder.

Why: The package is already installed on the laboratory PC, in a shared folder.

Fix: Cancel. Help → **Updater Settings** → Repository Folder: `C:/Users/Public/STM32Cube/Repository` → OK → restart CubeMX.

11. You browsed *into* the package folder

What you did: In Updater Settings you browsed down to `...\Repository\STM32Cube_FW_F4_V1.28.2`.

What you see: CubeMX still offers to download the package.

Why: The setting expects the folder that **contains** the packages, not one package.

Fix: Set it to `C:/Users/Public/STM32Cube/Repository` — type it, do not browse into a subfolder.

Building and running

12. You used the status bar Build button

What you did: Pressed the **Build** button in the blue status bar, or ran *CMake: Build* from the Command Palette.

What you see: `STM32F401RETx_FLASH.ld:56: syntax error` and no `.elf`.

Why: CubeMX has a known bug: the linker script it generates loses its memory region names. The build task in `.vscode` repairs the file first. The status bar button runs CMake directly and skips that repair. CubeMX breaks the file again on every regeneration.

Fix: Always build with **Ctrl+Shift+B** (Terminal → Run Build Task).

13. You picked the Release preset

What you did: When CMake Tools asked for a configure preset, chose **Release** or **RelWithDebInfo**.

What you see: The build succeeds, but debugging reports the `.elf` does not exist — the debug configuration looks in `build\Debug\`.

Fix: Choose **Debug**. To change it: Command Palette → *CMake: Select Configure Preset* → Debug, then Ctrl+Shift+B.

14. You pressed the in the editor title bar

What you did: Clicked the play triangle at the top right of the `main.c` editor, or chose *Run C/C++ File*.

What you see: A prompt to choose a debugger, or compiler errors mentioning `cl.exe`, `g++`, or missing standard headers.

Why: That button tries to compile and run the single file **on the PC**. Your program is for the STM32, not for Windows.

Fix: Press **F5**, or use Run and Debug view → next to **Debug (ST-LINK)**. F5 builds, flashes and starts debugging in one step; there is no separate "program the board" button.

15. Something else is holding the ST-LINK

What you did: Left a debug session running, opened a second VS Code window on the same project, or left STM32CubeProgrammer connected.

What you see: ST-LINK GDB server errors, "cannot connect to target", or the debugger starting and dying immediately.

Why: Only one program can use the ST-LINK at a time.

Fix: Stop the previous debug session (Shift+F5), close extra windows and CubeProgrammer, unplug and replug the board, then F5 again. The first F5 after replugging sometimes fails on its own; a second F5 usually works.

16. You answered the "install cube bundle(s)" notification

What you did: Clicked the notification that offers to install ST bundles.

What you see: F5 waits and never starts the debugger.

Why: The extension blocks debugging until the question is answered, and the lab PCs already have everything.

Fix: Answer **No**, or dismiss it. If F5 is stuck, check the bell icon in the bottom right for an unanswered question.

17. You added a `.c` file but not to `CMakeLists.txt`

What you did: Put a new file such as `sensor_math.c` in `Core\Src` and called its functions from `main.c`.

What you see: The build stops at the link step with `undefined reference to 'clamp_u16'`.

Why: A CubeMX CMake project compiles only the files listed for it. CubeMX lists its own files; your new files are not on any list yet. The compiler saw the declaration in the header, but the linker found no definition.

Fix: Add the file to `target_sources` in `CMakeLists.txt`:

```
target_sources(${CMAKE_PROJECT_NAME} PRIVATE
    # Add user sources here
    Core/Src/sensor_math.c
)
```

New header files in `Core\Inc` need no change. CubeMX never rewrites this part of `CMakeLists.txt`, so your lines stay after regeneration.

Keeping your work

18. You wrote code outside the USER CODE regions

What you did: Typed statements between `/* USER CODE END WHILE */` and `/* USER CODE BEGIN 3 */`, or anywhere outside a marked region.

What you see: The code works, until you change a pin in CubeMX and regenerate — then it is gone.

Why: Regeneration rewrites every line except those inside `USER CODE BEGIN … END` pairs.

Fix: Keep every statement you write inside a region. `BEGIN 2` for variables before the loop, `BEGIN 3` for the loop body.

19. You worked directly on `H:`

What you did: Opened the project from `H:\EE_Labs` and built it there.

What you see: Very slow builds, strange path errors, and sometimes a corrupted `build` folder.

Fix: Copy the project from `H:\EE_Labs` into `C:\Work\<studentnumber>` first, work there, and at the end of the session run the **Save to H** desktop shortcut. Anything left only on `C:` may be gone next week.

Warning: *Save to H* makes `H:\EE_Labs` an exact copy of `C:\Work\<studentnumber>`. Projects that are on `H:` but not on this PC are **deleted**. So at the start of every session copy **all** your projects back from `H:` — not only the one you are working on.

When you still need help

Bring these four lines, written down:

```
Expected:
Observed:
Evidence collected:
Smallest next test:
```